Introduction to Data Science, Assignment 2 by Group 110

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1 Question 1: Market Basket Analysis

1.1 a

The values we obtained from our code can be seen in table 1:

Metric	**Value**
Orders needed for 1% support	649
Mean support count per product	21.59
Median support count per product	4.0

Table 1: Support Count Statistics for Products

1.2 b

The top 10 products along with their absolute and relative frequency can be found in table 2:

Rank	**Product ID**	**Absolute Frequency**	**Relative Frequency**
1	24852	9376	0.14
2	13176	7701	0.12
3	21137	5480	0.08
4	21903	4818	0.07
5	47626	4096	0.06
6	47766	3670	0.06
7	47209	3605	0.06
8	16797	3199	0.05
9	26209	3017	0.05
10	27966	2728	0.04

Table 2: Top 10 Most Frequent Products with Rankings

Considering only this list, the maximum support a set of three items could receive would be 0.08, if every order that included product 21137 also included products 13176 and 24852.

1.3 c

The three most commonly bought items can be found in table 1.3:

Rank	Product ID	Product Name	Absolute Frequency
1	24852	Banana	9376
2	13176	Bag of Organic Bananas	7701
3	21137	Organic Strawberries	5480

Table 3: Top 3 Most Frequently Purchased Products

The three items only occur together in three orders (2275990, 850056 and 3218207). This is to be expected, as they include two variations of bananas (organic and non-organic). As customers typically buy only one variant of such a common product, buying two distinct variants is not expected to occur often.

1.4 d

The plot of number of sold products[1] per category can be found in figure 1.4. In figure 1.4, we plotted the mean and median number of items per order for each category.

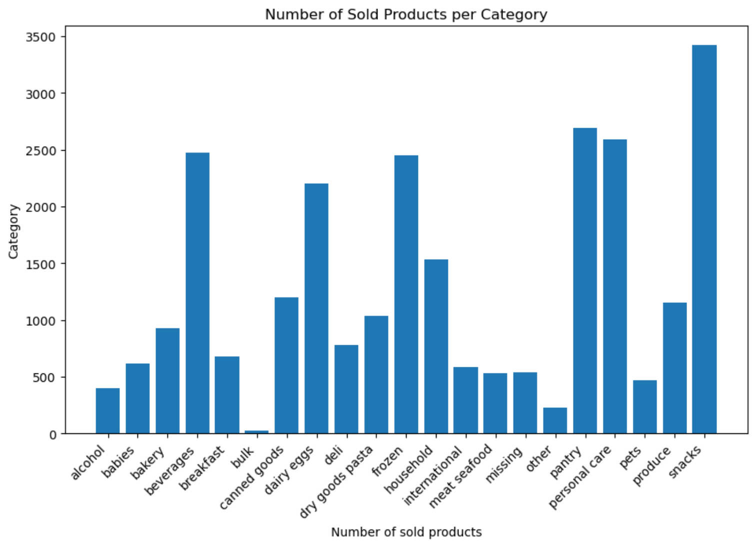

Figure 1: Number of sold products per category

1.5 e

The values of mean and median for the beverage category can be explained as follows: Most costumers purchase one beverage per order. However, there are

[1] We interpret "number of sold products" as the number of distinct products sold in each category (counting each product only once), instead of the total number of product sales per category.

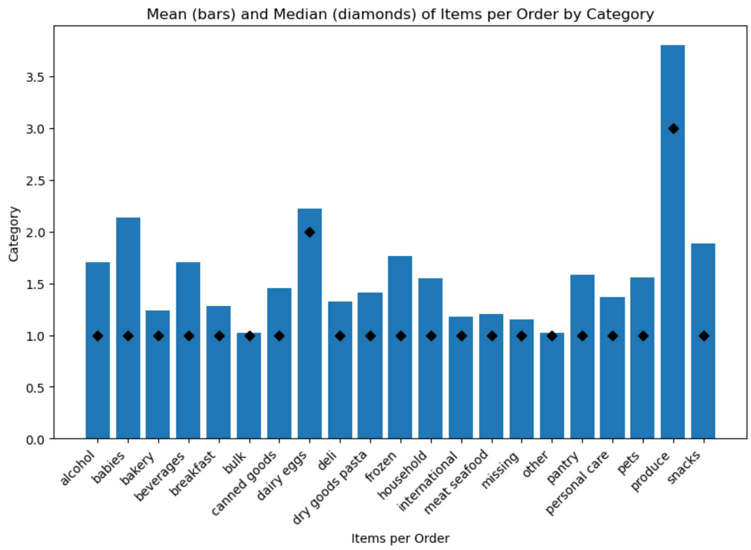

Figure 2: Mean and median of number of items per order per category

some orders that order significantly more beverages, bringing the average to 1.5 beverages per order.

1.6 f

- The names of the products in the largest frequent itemset can be found in table 4.
- There are 1017 frequent itemsets of size 2 or larger (relative frequency of 0.51).
- The most common item in the sets is 24852 (Banana), appearing in 138 sets.
- 1021 items only occur in a single set.

Product ID	**Name**
13176	Bag of Organic Bananas
47209	Organic Hass Avocado
21137	Organic Strawberries
27966	Organic Raspberries

Table 4: Products in the largest frequent itemset.

1.7 g

The plot of categories of most frequent itemsets can be found in figure 3:

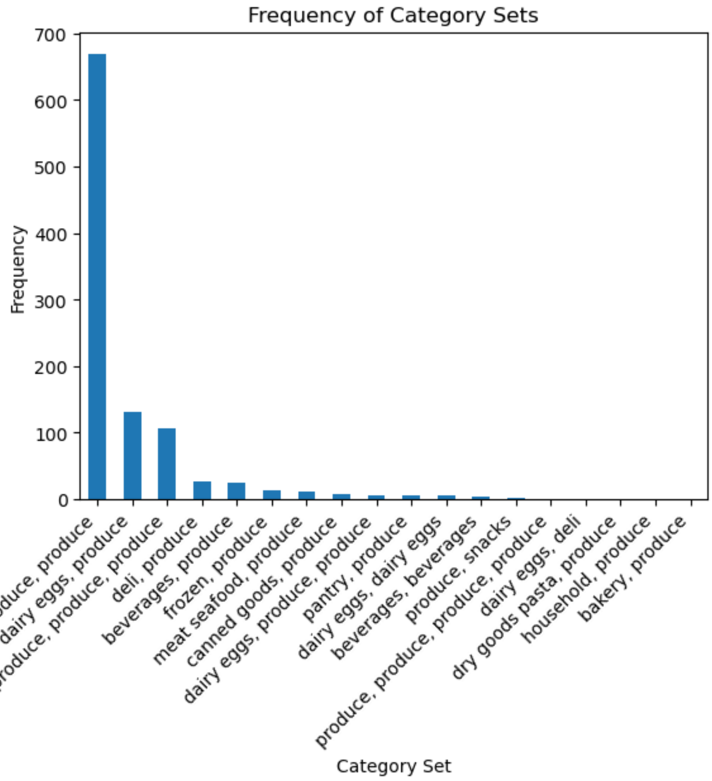

Figure 3: Frequency of category sets

1.8 h

The frequency of purchases with only produce items is not unexpected as most frequently bought items from the store are of the produce category. While the amount of products offered in this category is lower, purchases have to concentrate on few items. As most customers will buy some produce as a staple product in their regular shopping, this will lead to a lot of frequently bought together items coming from produce.

1.9 i

- The lift statistics can be found in table 5.
- There are 222 rules with at least one of the top 10 products in the antecedents.
- There are 421 rules with at least one of the top 10 products in the consequents.

Metric	Value
Mean lift	3.96
Median lift	2.82

Table 5: Statistics for lift values in association rules.

1.10 j

The rule with the highest confidence (that includes a top 10 product) is provided in table 6:

Antecedents	Consequents	Confidence
Organic Hass Avocado, Organic Strawberries, Organic Raspberries	Bag of Organic Bananas	0.62

Table 6: Association rule with the highest confidence.

When a customer buys an organic avocado, organic strawberries and organic raspberries, they are very likely to also buy organic bananas (in around 62% of cases). It is not surprising to see the most frequently bought item (organic bananas) in the consequents, as it is part of a large amount of purchases. This means that it is very broadly distributed among purchases and any association of buying these bananas with other products will have less confidence.

1.11 k

The mean and median values for lift between rules with and without the top 10 products can be found in table 7:

Metric	Without Top 10 Products	With Top 10 Products
Mean lift	18.05	3.10
Median lift	8.45	2.77

Table 7: Lift statistics comparison between niche rules with and without top 10 products.

The lift is significantly higher for rules without the top ten products. This is not surprising as their base frequency is low, meaning the denominator in the lift equation is very small, leading to high lift. As these items are rarely purchased, any association between them are likely to result in high levels of lift.

1.12 l

The rules that are both symmetrical and meet the required threshold can be found in table 8:

Product A	Product B	supportCount(a,b)	supportCount(b,a)
13176	21137	1149	404
33754	4957	57	53
36865	28465	61	59
33754	33787	69	47

Table 8: Symmetrical rules with support meeting threshold (Sequence analysis).

1.13 m

The confidence is calculated as support($A \cup B$)/support(A) and lift is calculated as support($A \cup B$)/(support(A) $\times$ support(B)). To satisfy a lift smaller than 1, support($A \cup B$) must be smaller than (support(A) $\times$ support(B)) and to satisfy a confidence greater than 0.2, support($A \cup B$) must be greater than $0.2 \times$ support(A). Substituting the lower bound into the lift inequality gives:

$$0.2 \times \text{support}(A) < \text{support}(A) \times \text{support}(B)$$

which simplifies to $0.2 <$ support(B). However, from answer (b), the highest relative frequency is only 0.14, so it is not possible to find an association rule with these metrics for this dataset.

2 Question 2

2.1 a

The petri net that was discovered using the inductive miner can be seen in figure 4:

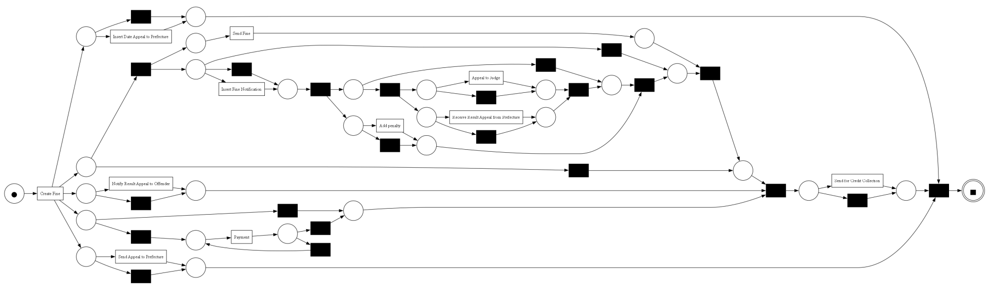

Figure 4: The petri net discovered in task 2a

Based on this model, we can make the following observations:

1. All traces have to start with Create fine.
2. The only activity that has to be executed before Appeal to Judge can be executed is Create fine (as all other activities can be skipped via silent transitions).
3. Only the Payment activity is part of a loop and can therefore be executed multiple times.

4. The model allows for a path that involves the Credit Collection to be involved without the Send Fine activity to be executed.

2.2 b

The number of occurences described can be found in table 9:

Case Description	Count
Go to collections after payment	1013
Add a fine before being sent out	0
Appeal to anyone after payment sent	10
Notification of result without appeal	5

Table 9: Frequency of specific process paths in case management.

2.3 c

The plot of cumulative variant frequency can be found in figure 5, the most common variants can be found in table 10:

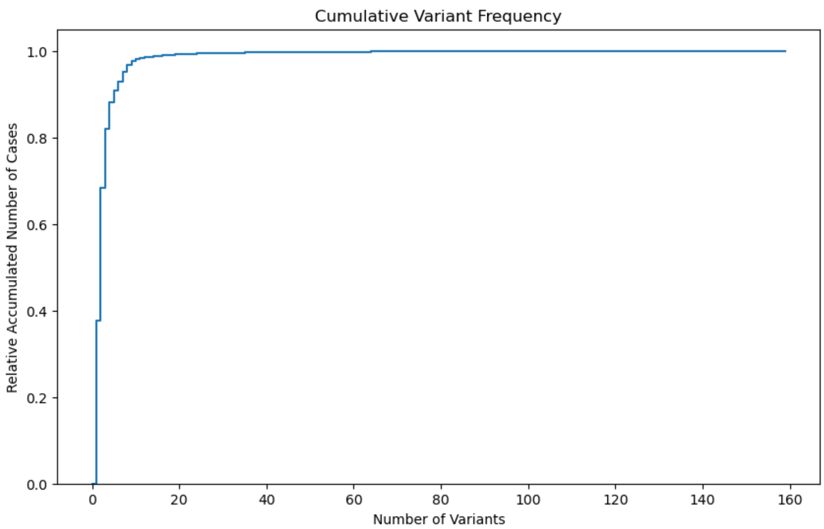

Figure 5: The cumulative variant frequency plot

Interpretation: The cumulative variant frequency chart shows a very steep increase for the first few variants, indicating that a small number of variants covers a large proportion of the cases. After roughly the first 10 variants, the curve becomes almost horizontal, meaning that many additional variants contribute only marginally to the overall case coverage.

2.4 d

The pie chart of case statuses can be found in figure 6:

Variant	Count
Create Fine → Send Fine → Insert Fine Notification → Add penalty → Send for Credit Collection	26872
Create Fine → Payment	22078
Create Fine → Send Fine	9631
Create Fine → Send Fine → Insert Fine Notification → Add penalty → Payment	4507
Create Fine → Send Fine → Insert Fine Notification → Add penalty → Payment → Payment	1806

Table 10: Top 5 Process Variants by Frequency

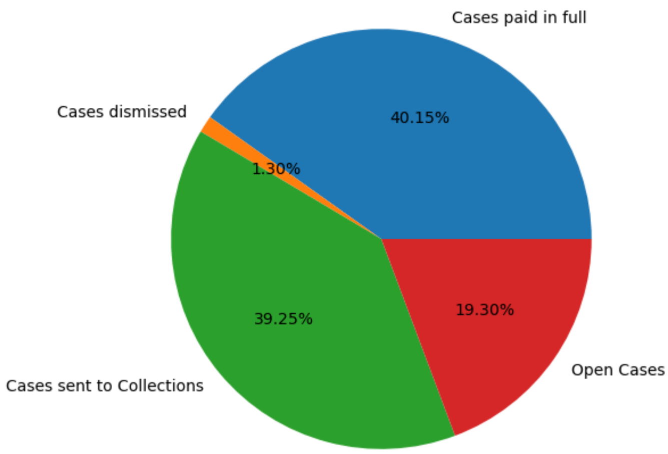

Figure 6: Pie chart of case resolutions

2.5 e

The discovered model can be found in figure 7:

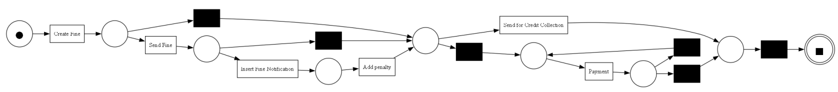

Figure 7: The model discovered from the most popular variants

The model discovered from the filtered log is significantly simpler than the model discovered from the full event log. After Create Fine, it mainly consists of two parts. The first part involves the activities Send Fine, Insert Fine Notification, and Add penalty, which may be executed sequentially or skipped. The second part concerns the resolution of the case and consists only of the Payment and Send for Credit Collection activities. In contrast, the model discovered from the full log includes many additional activities related to appeal procedures. Regarding the timing of payments, the model allows Payment to occur at different moments in the process: directly after Create Fine, after Add penalty, and in a self- repeating manner. This is consistent with the dominant behavior observed in the full log. In addition, the model also permits Payment to occur after Send Fine, which is not a dominant behavior observed in the full log but is a dominant behavior in the filtered log. Finally, Payment and Send for Credit Collection are in an exclusive choice relation: a case proceeds either with a payment or with sending it for credit collection, but not both. This indicates that payment and credit collection represent alternative ways of handling outstanding fines.

2.6 f

The log fitness measures (found in table 11), on average, how well the discovered model can reproduce the behavior observed in the event log, whereas the percentage of perfectly fitting traces counts only the traces that can be replayed without any deviation. In our case, the model discovered from the top five variants captures the dominant behavior well, which leads to a relatively high fitness value despite the presence of many deviations. However, many traces contain additional behavior that is not represented in the simplified model like activities related to appeal procedures, meaning they are not perfectly fitting and therefore cause the percentage of perfectly fitting traces to drop significantly.

Metric	**Value**
Perfectly fitting traces	80.26%
Overall log fitness value	0.97

Table 11: Fitness Statistics for the Process Model

3 Question 3

3.1 a

The word frequency histogram can be found in figure 8.

The following problems may influence this distribution:

- Common English stopwords (”and”, ”or”, ”is”, ”a” etc.) are not removed, leading to them dominating the list of most frequent words. This problem could be addressed by removing those words.
- Punctuation and capitalization may influence the results (”data,” and ”data” are counted as different words). This can be addressed by removing capitalization and punctuation.

3.2 b

The tokenized histogram for lecture 1 can be found in figure 9.

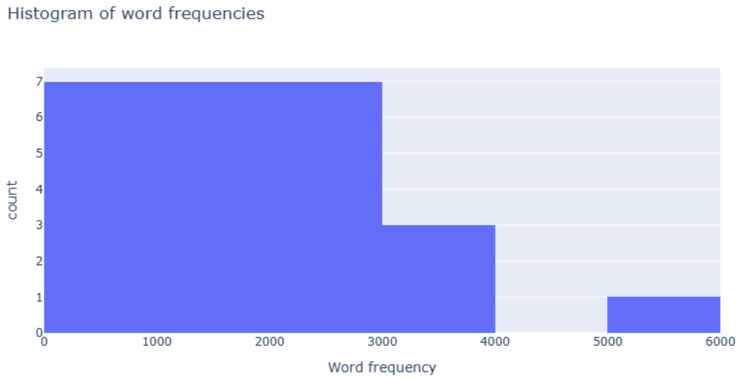

Figure 8: The histogram of word frequencies.

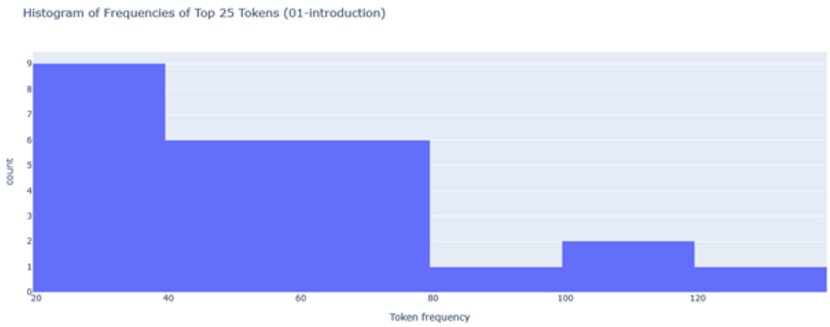

Figure 9: The histogram of tokenized word frequencies in lecture 1.

3.3 c

The stacked histogram can be found in figure 10. The token data appears in every lecture and dominates the distribution, with particularly high frequencies in the first three lectures. The token easily occurs relatively rarely but is evenly distributed across lectures. The remaining tokens are more lecture-specific, reflecting the lecture topics, with database appearing exclusively in lecture 10.

3.4 d

We generated the following outputs:

- For n=3: introduction to data science master and its not shown so again this other one is not possible and so on the part that is possible for specific tasks so the people that invented
- For n=4: introduction to data science so this lecture will be a lecture on regression they will be provided to you so the question earlier was like how to interpret these factors the different rate
- For n=5: introduction to data

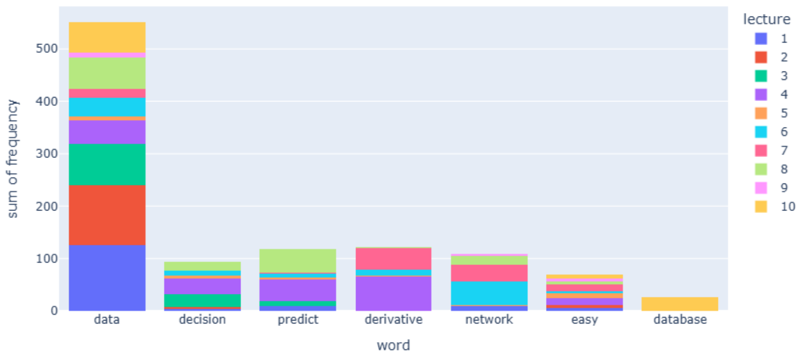

Figure 10: The stacked histogram of words across lectures.

- For n=2: introduction to data set on the or regression model prediction model prediction model prediction model prediction model prediction model prediction model prediction model prediction model prediction model prediction model prediction model prediction model prediction model

For n = 2, the generated text starts reasonably but quickly becomes repetitive due to the very short context, which strongly favors frequent word pairs and can lead to looping. For n = 3, the text is readable but often lacks clear semantic meaning, as the limited context allows many plausible continuations. For n = 4, the output remains readable and becomes more semantically meaningful because the model conditions on a longer context. For n = 5, no continuation is generated because generation requires the existence of an n-1 token context in the ConditionalFreqDist, and such longer contexts are much rarer. For $n > 5$, this sparseness problem becomes even more noticeable given the limited dataset, so we expect empty or very short outputs.

3.5 e

The result for `Gradient descent approach` can be found in table 12, the result for `Beer and Diapers` can be found in table 13. The system retrieves 04-regression as the top match with timestamps focusing on optimization concepts (error curves, parameter fixing), and 06-neural-networks-1 as second highlighting derivative calculations and weight updates which are both highly relevant to gradient descent in the lecture context. The retrieval identifies 10-frequent-itemsets as the top lecture, with timestamps directly covering the "beer and diapers" market basket example. The second match (01-introduction) is very weak and unrelated, demonstrating that the hierarchical approach effectively prioritizes the specific association-rule topic while assigning negligible scores to generic introductory content.

Lecture Name	Score	Timestep List (Score)
04-regression	0.08	[3324.14, 3353.17] (0.28), [1907.18, 1935.92] (0.28)
06-neural-networks-1	0.03	[2272.58, 2301.78] (0.45), [1921.86, 1952.67] (0.31)

Table 12: TF-IDF Search Results for Gradient Descent Approach

Lecture Name	Score	Timestep List (Score)
10-frequent-itemsets	0.03	[1521.46, 1549.01] (0.41), [857.25, 886.69] (0.34)
01-introduction	0.003	[3881.79, 3904.34] (0.23), [2.35, 31.36] (0.0)

Table 13: TF-IDF Search Results for Beer and Diapers

4 Question 4

4.1 a

4.1.1 i

The statistics obtained from the dataset can be found in table 14.

Metric	Value
Time range	1/2003 – 9/2023
Months tracked	249
Amount of flights	192,100,234

Table 14: Overview Statistics for the Flight Dataset

4.1.2 ii

The timeseries plot of total flights can be found in figure 11. The time series of monthly flight counts shows a clear seasonal pattern with recurring yearly peaks and troughs, as well as a long-term trend. A sharp and unprecedented drop occurs in early 2020, probably due to the COVID-19 pandemic, followed by a gradual recovery in subsequent months, although flight levels remain below pre-pandemic values.

4.1.3 iii

The timeseries plot of total flights against total passengers can be found in figure 12. The development of passengers follows a similar pattern to that of flights. However, it is visible that while a general downward trend in number of flights is seen, this is not present in the number of passengers. This indicates that demand for flights has not gone down, but plane size or utilization of planes has increased.

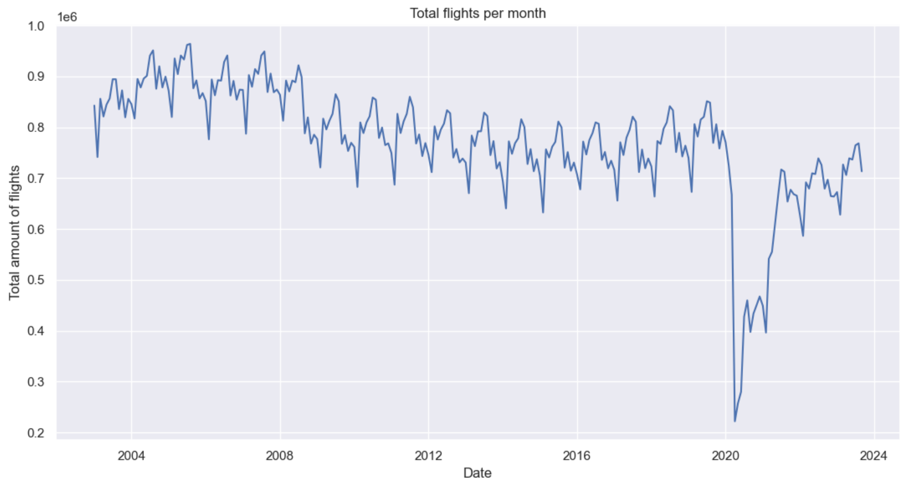

Figure 11: Timeseries plot of total flights per month

4.2 b

4.2.1 i

The plot of yearly seasonality for both passengers and flights can be found in figure 13. Both flights and passengers show a strong and consistent seasonal pattern in most years (2003-2019), with clear summer peaks (June-August, usually highest in July/August) and winter lows (January-February). The year 2020 strongly deviates due to the pandemic, with a sharp drop from March and almost no summer recovery. From 2021 onwards, the typical seasonal shape returns, though passenger volumes often show higher peaks than pre-2020 levels.

4.2.2 ii

The Pearson correlation of 0.57 shows only a moderate overall relationship between passengers and flights across the full period. This lower-than-expected value is consistent with subtask a.iii, because the strong pre-2020 similarity is heavily offset by the very different post-2020 recovery patterns (fast passenger rebound vs. slow/partial flight recovery), which weakens the linear association over the entire time series.

4.2.3 iii

The plots of three correlograms with no differencing, one differentiation and two differentiations applied can be found in figure 14. The original ACF (0 differencing) decays very slowly with significant positive values up to lag 24 and noticeable seasonal spikes every 12 months, indicating non-stationarity and strong yearly seasonality. After 1st differencing, most autocorrelation drops near zero except for prominent positive spikes at lags 12 and (weaker) 24. After 2nd differencing, the ACF is mostly insignificant except for the same clear spikes at lags 12 and 24. Lag 12 (and to a lesser extent lag 24) consistently show

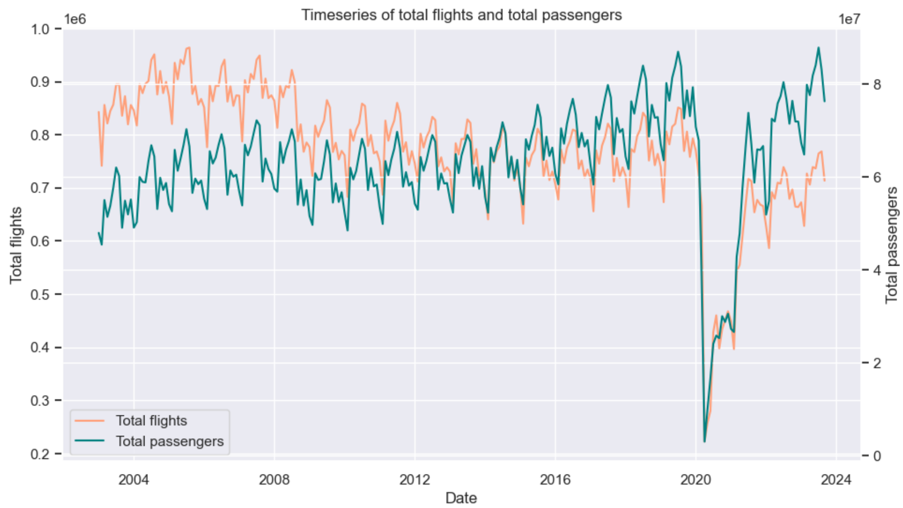

Figure 12: Timeseries plot of total flights per month against total passengers per month.

the strongest positive correlations across all three plots, reflecting the dominant annual seasonal cycle.

4.2.4 iv

The plotted stl decomposition can be found in figure 15. The STL decomposition (period=12) shows a gradually declining trend until 2019, a sharp drop in 2020 (likely caused by the COIVD-19 pandemic), and partial recovery afterward; a very regular seasonal component with summer peaks and winter troughs; and residuals that are mostly small except for large negative outliers in 2020 and some positive ones during recovery. The residual series is stationary. The ADF test statistic is -6.6518 with p-value 0.0000 (well below all critical values), strongly rejecting non-stationarity.

4.3 c

4.3.1 i

The flight and passenger time series are generally suitable for forecasting, as they exhibit stable and consistent patterns prior to 2020. The year 2020 represents an outlier due to a major external shock, introducing temporary noise into the series. However, this disruption can be accounted for or treated separately, allowing the series to remain useful for forecasting.

4.3.2 ii

We performed a grid search over the three parameters p, d, q, with $p \in [1, \dots, 9]$, $d \in [0, \dots, 3]$ and $q \in [0, \dots, 12]$, returning tuned parameters at the first result beating the naive approach in all error categories. This returned the configu-

Figure 13: Plot of yearly flights and passengers per month

ration $\{p = 1, d = 0, q = 0\}$ as the first superior configuration. The resulting values for this can be found in table 15.

Metric	Naive Forecaster	ARIMA Forecaster
RMSE	51192.10	43464.06
MAE	36663.86	33692.37
MAPE	0.05	0.04

Table 15: Performance Comparison between Naive and ARIMA Forecasters

4.3.3 iii

The requested plot can be found in figure 16. The forecast appears overly smooth and does not capture the variability of the test data, indicating that the model does not provide a good forecast. While it provides values close to a running average of values, its utility is highly questionable.

5 Question 5

5.1 a

The mathematical functions for our map and reduce functions are as follows:

- $\text{map}_1 \in (N_0 \times S) \rightarrow (C \times (T \times A))$
- $\text{reduce}_1 \in (C \times (T \times A)) \rightarrow (C \times A^*)$
- $\text{map}_2 \in (C \times A^*) \rightarrow ((A \times A) \times N_0)$
- $\text{reduce}_2 \in ((A \times A) \times N_0) \rightarrow ((A \times A) \times N_0)$

The computation of the Directly-Follows Graph (DFG) is divided into two MapReduce jobs. In the first job, the map function groups events by their case id. It takes the line number and the whole log line as input, extracts the case id, activity, and timestamp, and outputs the activity and timestamp group by the case id. The reduce function receives all activities and timestamps belonging to the same case, sorts them according to the timestamps, and outputs the ordered sequence of activities for each case. In the second job, the map function processes the sorted activity sequence of each case and emits all directly-follows activity pairs. Each pair is emitted with a count of 1. The reduce function groups identical activity pairs and sums their counts, producing the final DFG where each edge represents a directly-follows relation and its frequency.

5.2 b

The figure of the generated DFG can be found in figure 17. We generate the DFG as follows: First, we parse each log line and extract case id, activity, and timestamp, emitting (`case_id`, (timestamp, activity)). Next, groupByKey groups all events belonging to the same case. Then, mapValues sorts the events of each case by timestamp to obtain the ordered sequence of activities. After that, flatMap extracts all directly-follows relations by emitting each adjacent activity pair from the sequence with count 1. Finally, reduceByKey sums these counts per activity pair to count the occurrences of the directly-follows relations.

5.3 c

Adding a filter transformation removes directly-follows relations with a frequency smaller than 100. By applying filter(lambda x: x[1] ¿= 100), all arcs whose frequency is below 100 are filtered out, resulting in a simplified Directly-Follows Graph. The filtered DFG can be found in figure 18.

5.4 d

First, each log line is parsed to extract the activity and resource, emitting pairs of the form (activity,resource) (activity,resource). Next, groupByKey groups all events belonging to the same activity. Then, the reduce step counts how many times each resource executes the activity. Finally, for each activity, the resource that most frequently executes it is selected and emitted. The returned values can be found in table 16.

Activity Name	**Resource/ID**	**Frequency**
Create Fine	R-538	8608
Send Fine	MISSING	103987
Insert Fine Notification	MISSING	79860
Add penalty	MISSING	79860
Send for Credit Collection	MISSING	59013
Payment	MISSING	77601
Insert Date Appeal to Prefecture	MISSING	4188
Send Appeal to Prefecture	MISSING	4141
Receive Result Appeal from Prefecture	MISSING	999
Notify Result Appeal to Offender	MISSING	896
Appeal to Judge	R-0	555

Table 16: Frequency and Resource Attribution per Activity

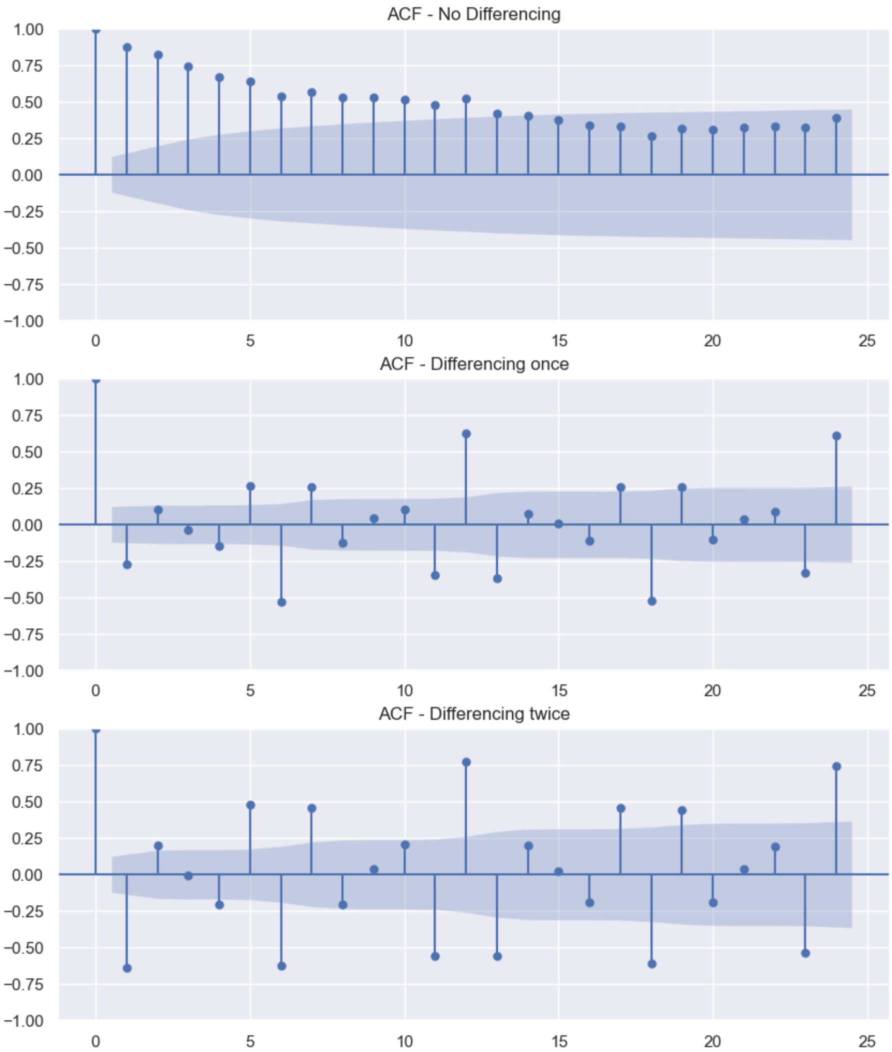

Figure 14: Plot of the three correlograms.

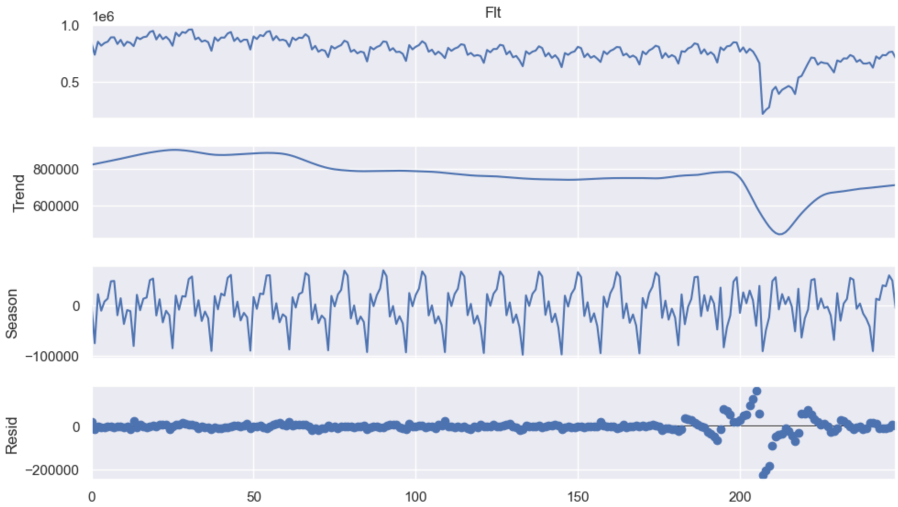

Figure 15: Plot of the STL decomposition.

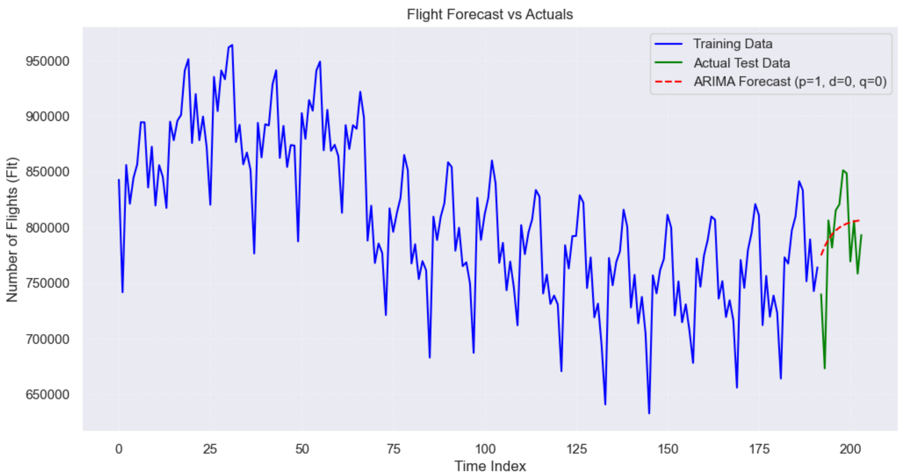

Figure 16: Plot of the three stl decomposition.

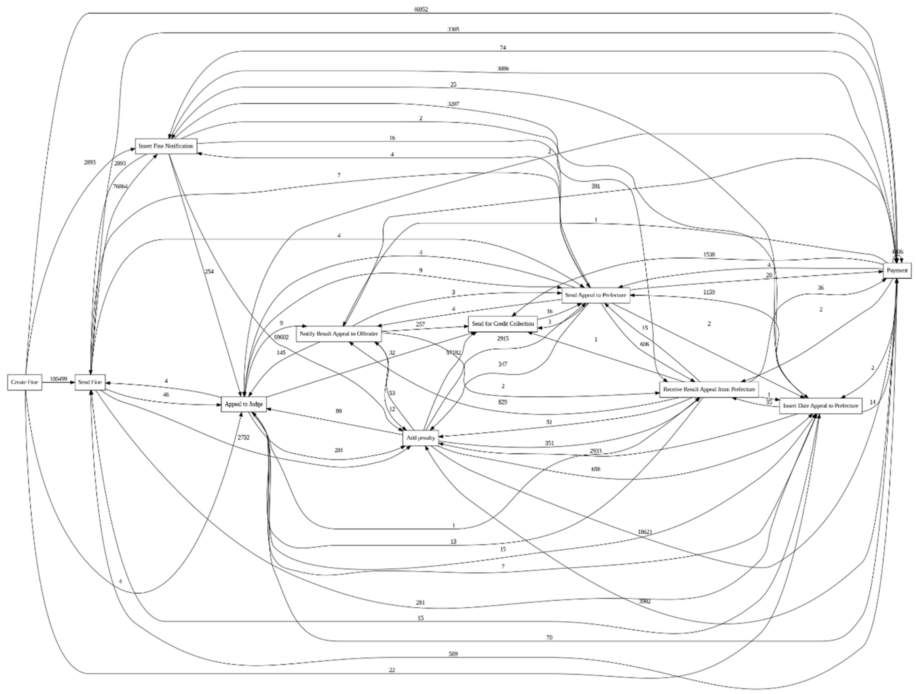

Figure 17: DFG generated from the event log.

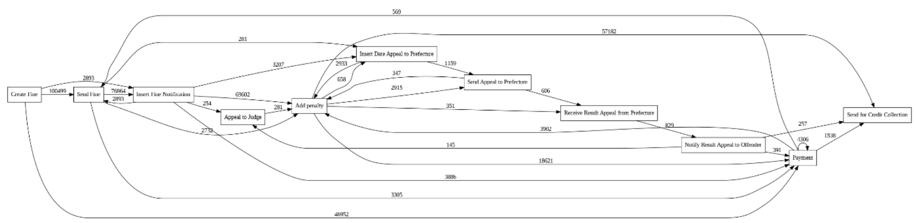

Figure 18: The filtered DFG generated from the event log.